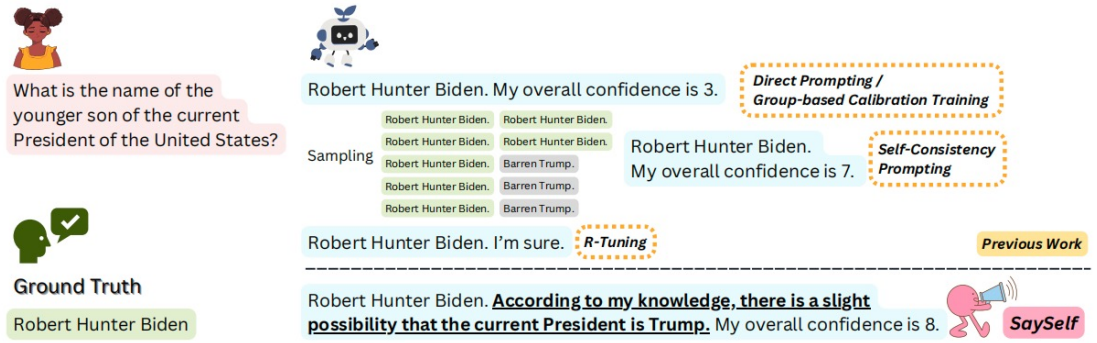
SaySelf

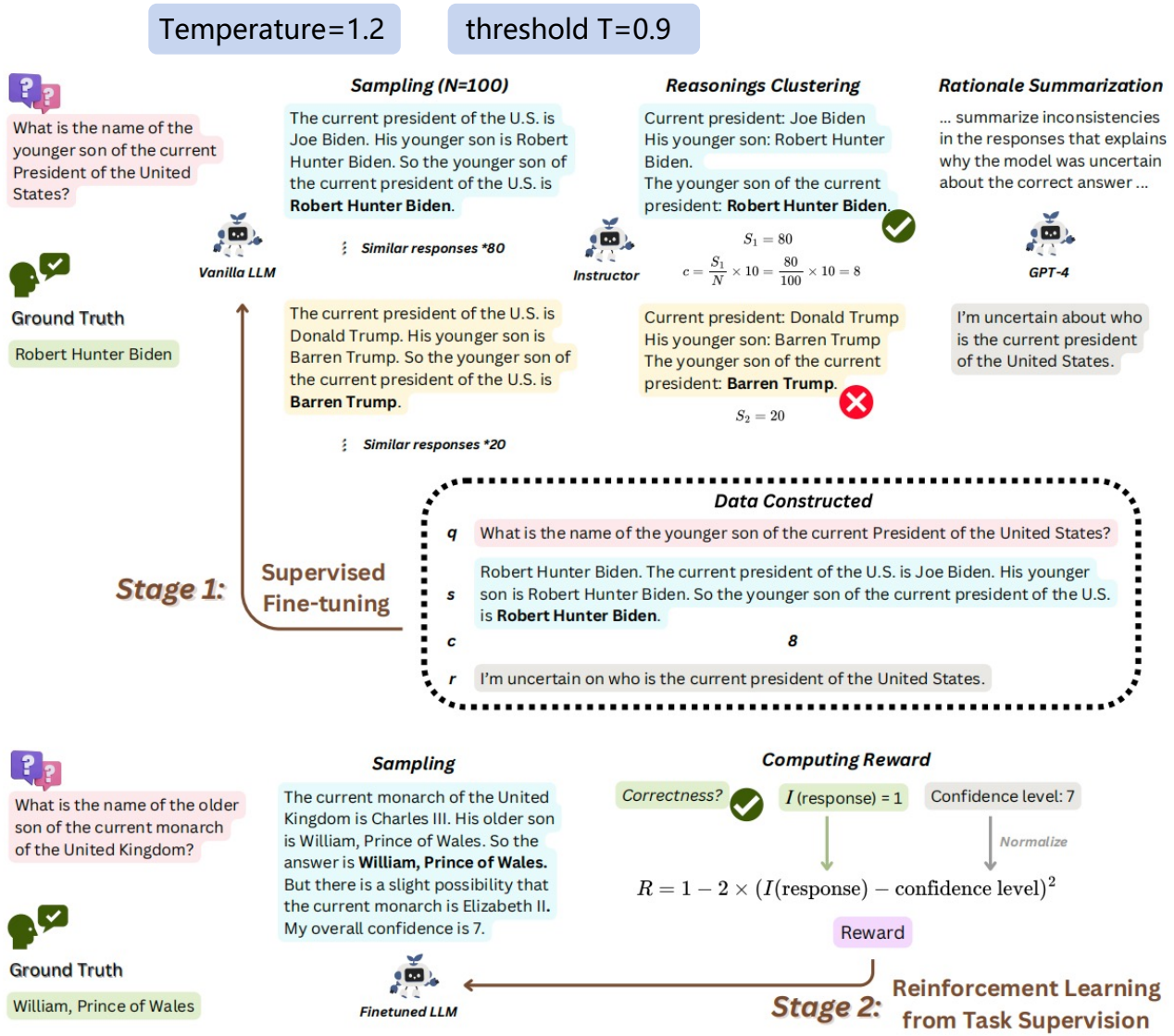
SaySelf[113]



☐ Overview

SaySelf trains LLMs in **two stages**:

- **Stage 1 – Supervised Fine-tuning :**
  1. LLM generates multiple answers for the same question and clusters them.
  2. GPT-4 summarizes uncertainty causes (the "self-reflective rationale").
  3. The model learns to output answer + rationale + confidence level.
- **Stage 2 – Reinforcement Learning :**
  1. Confidence values are refined to align predicted confidence with empirical correctness.



[113] SaySelf: Teaching LLMs to Express Confidence with Self-Reflective Rationales, arxiv 2024

SaySelf[113]

- **Task Type:** Open-ended generation

- **Introduction:** SaySelf introduces a self-reflective framework that enables LLMs to express their own confidence levels. Through a two-stage process involving SFT and PPO RL, the model learns to generate task responses, rationales and confidence scores, improving both confidence calibration and interpretability.

Type	Uncertainty Sources	Model Accessibility	Training Necessity	Consumption	Evaluation
Single-pass	Reasoning	White-box	Fine-tuning	High	Calibration

[113] SaySelf: Teaching LLMs to Express Confidence with Self-Reflective Rationales, arxiv 2024